# Hybrid Time Series Forecasting of Apple Stock Prices Using Mini-Transformer and LSTM Models

Romington Hydro
*Computer Science Department*
*School of Computer Science*
*Bina Nusantara University*
Jakarta, Indonesia 11480
romington.hydro@binus.ac.id

Zefanya Delvin Sulistiya
*Computer Science Department*
*School of Computer Science*
*Bina Nusantara University*
*Jakarta, Indonesia 11480*
*zefanya.sulistiya@binus.ac.id*

Alexander Agung Gunawan Santoso
*Computer Science Department*
*School of Computer Science*
*Bina Nusantara University*
*Jakarta, Indonesia 11480*
*aagung@binus.edu*

***Abstract*—Forecasting stock prices requires models capable of navigating complex, nonlinear patterns under volatile market conditions. This research proposes a hybrid deep learning frame- work that integrates Mini Transformer and LSTM architecture to predict Apple Inc.'s daily closing prices. Historical stock data were collected via the yfinance library and preprocessed using scaling and sliding window techniques to create sequences. Each model used in this research (Mini Transformer, LSTM, and the proposed hybrid) was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE). Results shows that the hybrid model achieves superior predictive accuracy, with an MAPE of 1.6%, MAE of 2.4847 and an MSE of 10.4762, outperforming standalone method approaches. While the Mini Transformer and LSTM models each capture temporal features to some degree, the hybrid design leverages global attention mechanisms along with recurrent layers, resulting in a more robust forecasts. Insights from this research suggest that combining self-attention with recurrent memory cells can help mitigate the noise and structural shifts in financial time series. Future research could incorporate more complex market signals and larger-scale Transformers to further refine predictive performance in real-world trading applications.**

***Keywords— Time-Series Forecasting, LSTM, Transformer, Hybrid Models, Financial Data***

## I. Introduction

Stock price forecasting remains a foundational element of financial markets, influencing investment strategies, risk management, and economic stability. With global equity markets exceeding $100 trillion in capitalization [1], even marginal improvements in forecasting accuracy can yield substantial economic value. Apple Inc. (NASDAQ:AAPL), as the world's most valuable publicly traded company with a market capitalization exceeding $3 trillion as of 2024, presents a compelling case study due to its market dominance, technological innovation cycles, and sensitivity to macroeconomic factors.

The challenge of predicting stock prices comes from the complex interplay of multiple factors, including market sentiment, geopolitical events, and inherent characteristics of financial time series such as non-stationarity, high volatility, and noise [2]. Traditional statistical approaches such as autoregressive integrated moving average (ARIMA) models, while theoretically sound for stationary data, often fail to capture the non-linear dynamics and structural breaks prevalent in modern financial markets. This limitation becomes particularly apparent in technology stocks like Apple, where product launch cycles and supply chain disruptions can create sudden price discontinuities [3].

The advancement of deep learning has revolutionized financial time series forecasting, with Long Short-Term Memory (LSTM) networks emerging as a dominant architecture since their introduction by Hochreiter and Schmidhuber. LSTMs address the vanishing gradient problem through their gated memory cells, enabling effective modeling of long-range dependencies in stock price sequences [4]. Recent studies demonstrate LSTMs achieving 18-25% improvement in directional accuracy over ARIMA for NASDAQ-listed stocks [5]. However, the sequential processing nature of LSTMs limits their ability to simultaneously model multiple temporal resolutions and global dependencies.

Transformers, initially developed for natural language processing, have recently demonstrated strong potential in financial forecasting by leveraging their self-attention mechanisms to process entire sequences in parallel, identifying cross- temporal relationships effectively [6]. However, despite their advantages, Transformers face challenges in financial applications, including quadratic computational complexity and sensitivity to high-frequency market noise. Hybrid approaches aim to mitigate these challenges by enhancing accuracy and robustness in real-world scenarios [7].

Despite recent advancements, most current models tend to concentrate either on local temporal dependencies, as seen in LSTMs, or on global relationships, as in Transformers, thereby not fully leveraging their combined advantages. Although earlier hybrid models have demonstrated improved performance, they frequently encounter high computational costs or struggle with generalization to noisy, real-world financial data. This paper aims to fill this gap by presenting a new hybrid architecture that merges the lightweight Mini-Transformer with LSTM, creating a balanced design that enhances both local and global sequence modeling while ensuring computational efficiency. Our contribution is the development of an architecture that effectively integrates attention mechanisms with sequential memory, specifically tailored for stock price forecasting tasks like those of Apple, which exhibit structural breaks, volatility, and nonlinear dependencies. By systematically applying dropout, sequence windowing, and MinMax scaling, we enhance generalization and mitigate overfitting, providing a streamlined solution that surpasses previous hybrid models in terms of predictive accuracy and practical usability.

Hybrid models combining the temporal modeling strengths of LSTMs with the contextual awareness of Transformers have emerged as a promising solution. For instance, [8] proposed a Transformer-LSTM hybrid framework, achieving improved accuracy by integrating attention mechanisms with LSTM's capacity for time-series

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forecasting. Similarly, [9] demonstrated that an empirical mode decomposition-enhanced LSTM model outperformed conventional LSTM approaches in financial prediction tasks by achieving a 39% improvement in forecasting accuracy.

For stock market forecasting, advanced hybrid approaches like the ARFIMA-LSTM model successfully reduced mean absolute percentage error (MAPE) by incorporating statistical models and LSTMs to manage financial data's nonlinear and chaotic nature [10]. Furthermore, a CNN-LSTM-Transformer, hybrid model demonstrated superior accuracy in handling long-term dependencies and seasonal patterns in time-series data [11].

This study addresses the persistent challenges in stock price forecasting: notably high volatility, nonlinear dependencies, and sensitivity to market noise, by proposing a hybrid architecture that integrates the complementary strengths of Mini-Transformer and LSTM networks. Although traditional approaches struggle to balance global pattern recognition with localized temporal dynamics, our framework leverages the Transformer's parallelized self-attention to model cross- sequence relationships and the sequential processing of LSTM to capture fine-grained temporal dependencies. To mitigate overfitting and improve generalization, the model incorporates dropout regularization, sequence windowing, and MinMax scaling, while maintaining computational efficiency through optimized attention mechanisms. By unifying these components, the hybrid design addresses critical limitations of standalone architectures, offering a robust solution that adapts to structural shifts inherent in financial time series. This work advances hybrid model design for stock forecasting, demonstrating how synergistic integration of attention-based and recurrent networks can improve predictive accuracy and practical utility in noisy, real-world market environments.

## II. Literature Review

In time series forecasting domain, there are a large number of theories and approaches on how to do time series forecasting. The methods previously proposed ranges from foundational statistical approaches to advance deep learning methods. Researchers have increasingly turned to deep learning architectures to address these complexities, with LSTM- based models emerging as a dominant approach due to their ability to model temporal dependencies. Early work by Shi et al. [12] demonstrated LSTM's superiority over traditional methods like ridge regression for Apple's stock forecasting, achieving a 12% reduction in RMSE by capturing nonlinear price trends. Subsequent studies expanded LSTM's utility through feature engineering: Chaudhary et al. [13] integrated sentiment analysis from news articles, improving directional accuracy by 15% for Apple and Tesla stocks, while Babu and Sathyanarayana [14] incorporated technical indicators such as RSI and moving average to enhance fast trading predictions, reaching 92.3% accuracy. Bidirectional LSTM variants further advanced the field, with Han and Fu [15] modeling past-future dependencies in high-frequency NASDAQ data to reduce multi-step forecast errors by 18%.

The rise of Transformer-based models has introduced new opportunities for global dependency modeling in time-series tasks. Though initially designed for NLP, Transformers have been adapted to financial forecasting through scaled-down architectures like Mini Transformers. For instance, Ding [16] fused CNNs with LSTM in a hybrid framework, using convolutional layers to extract localized features (e.g., price spikes) and LSTM to capture long-term trends, achieving a 22% RMSE improvement over standalone models. Meanwhile, Wu [17] combined linear regression with LSTM (LR-LSTM) to predict Apple's stock, leveraging technical indicators to attain an RMSE of 2.14. Recent innovations like the Temporal Fusion Transformer (TFT) have further bridged gaps between NLP and finance: Niu et al. [18] applied TFT to global indices, combining self-attention with sequence-to-sequence LSTMs to outperform DeepAR by 14% in multi-horizon peak predictions.

Hybrid models dominate the recent literature, merging complementary techniques to address the limitations of standalone architectures. Kim and Won [19] pioneered this trend with an LSTM-GARCH hybrid, modeling volatility clustering in stock indices to reduce prediction intervals by 30%. Similarly, Bukhari et al. [10] integrated ARFIMA with LSTM to handle fractional integration in S&P 500 data, achieving an MAE of 0.02694. Recent studies have expanded hybrid designs to include decomposition techniques and optimization algorithms: Kaur et al. [20] paired LSTM with linear regression to balance temporal and linear relationships in Apple's stock, achieving 89.7% accuracy, while Chen et al. [21] introduced a Beluga Whale Optimization-guided Informer-LSTM model, reducing training time by 30% without sacrificing performance (RMSE: 3.02). Signal decomposition methods like Variational Mode Decomposition (VMD) have also gained traction, with Niu et al. [18] decomposing stock data into intrinsic mode functions (IMFs) to isolate noise and improve LSTM's robustness, yielding a 39% MAPE reduction.

Emerging trends emphasize multi-modal data integration and real-time adaptability. Teixeira and Barbosa [22] demonstrated that models like GRU and XGBoost outperform pure LSTM in high-volatility scenarios, with GRU-LSTM hybrids achieving a 1.8% MAPE for Apple's stock. Despite these advancements, recent critiques by Elsayed et al. [23] question the necessity of complex deep learning models for time-series forecasting, demonstrating that simpler statistical approaches often match or exceed neural networks in long-term predictions. This aligns with findings by Alamu and Siam [24], where ARIMA outperformed LSTM for 10-year Apple stock forecasts (MAPE: 2.1%), underscoring the need for context- aware model selection. Recent efforts address this through architectures like online learning LSTMs with incremental updates [17], dynamically adapting to concept drift in volatile markets.

This synthesis highlights the field's progression from standalone models to interdisciplinary hybrid frameworks while identifying persistent gaps in scalability and interpretability challenges our work addresses through a streamlined Mini Transformer-LSTM architecture optimized for Apple's stock.

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## III. RESEARCH METHODOLGY

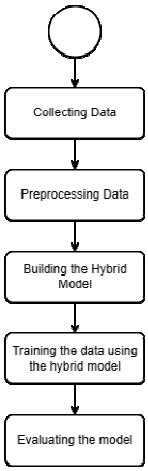


Fig. 1. Research Workflow

The process to conduct this research for this study are shown in the workflow in Fig.1. Gathering the experiment's data in the initial step. Once the dataset has been gathered, the data is preprocessed before pass to the model to increase the accuracy.

### A. Dataset

For the dataset used in this research, the objective was to collect historical stock data from Apple Inc. The data was fetched using the yfinance library, which are easy to access financial data from Yahoo Finance. Specifically, the closing prices of Apple's stock were retrieved over a specified time period, capturing the fluctuations in the stock's value after accounting for corporate actions such as dividends and stock splits.

The raw time-series data are stored in a variable of dataframe, ensuring an organized format suitable for analyzing and further processing. This dataset provides a foundation for exploring trends, volatility, and other financial metrics related to Apple's stock performance.

### B. Preprocessing

The 'Close' column is used when doing a forecast, as it indicates how well the stock actually performed, taking into consideration company activities such as dividends and stock splits. Similarly, so that the model performs better, the prices of the stocks are placed between 0 and 1 using the MinMaxScaler. This guarantees that all input features are at the same level, which keeps any single feature from making it easier or harder for the model to learn to an unreasonable degree [22].

The next step is creating a sequence that will be implemented to generate sequences of stock price for model training. This step takes the scaled data price and a specified sequence length as inputs, creating feature sequences of the 60 time steps is used, which allows the model to learn from historical patterns over a two-month period.

The window size refers to the number of time steps in each input sequence that will be used by the model to forecast. In this study, the window size is set to 60 means that the model will consider 60 consecutive data points as an input. The horizon is set to 1, indicating that the model predicts the next immediate value following the 60 time steps. This will effectively structure time series data to train the models, allowing the model to predict better by leveraging historical data [26].

### C. Proposed Method

The hybrid Transformer-LSTM model architecture follows a sophisticated neural network design for sequence processing. The input data first traverses through multiple Transformer encoder blocks, which utilize multi-head attention mechanisms to capture complex contextual relationships. These blocks simultaneously process input features through parallel attention and feed-forward neural network layers, enabling sophisticated feature extraction [27].

In parallel, the original input is directly fed into an LSTM layer, allowing the model to capture sequential dependencies and temporal dynamics. After processing, both the Transformer and LSTM paths undergo global average pooling and dropout regularization, which helps prevent overfitting and reduces computational complexity [28].

The final stage involves a dense output layer that consolidates the features extracted from both Transformer and LSTM pathways. This hybrid architecture combines the strengths of self-attention mechanisms (Transformers) with recurrent neural network capabilities (LSTM), enabling the model to effectively learn intricate patterns in sequential data while maintaining robust generalization performance.

By integrating these complementary neural network architectures, the model can capture both global contextual relationships and sequential temporal dependencies, making it particularly powerful for complex sequence modeling tasks across various research domains.

The proposed method is visualized in Fig. 2.

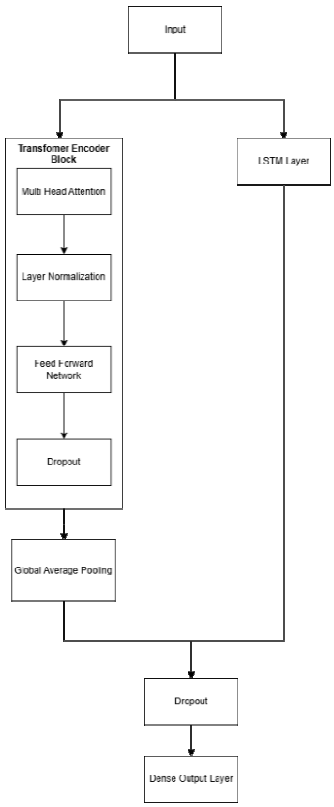


Fig. 2. The Hybrid Model Architecture

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### D. Evaluation Metrics

The model is assessed with 2 common metrics in regression which is Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The corresponding equations for each loss function are calculated as follows:

- MAE reflects the average absolute deviation of prediction values from the actual values, disregarding the error's directions.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (1)$$

- MSE calculates prediction errors by squaring it, which highlights larger errors more than the smaller ones due to squaring.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (2)$$

- MAPE expresses the error percentage between the actual values and predicted values [29]

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{A_i - F_i}{A_i}\right| \times 100 \quad (3)$$

## IV. Result and Discussions

This Research uses Google Colab to run the model. Here are the results of the experiment:

TABLE I
EXPERIMENTAL RESULT

| Model | Evaluation Metrics | | |
|---|---|---|---|
| | MAE | MSE | MAPE |
| Proposed Method | 2.4847 | 10.4762 | 1.61% |
| Mini Transformer | 14.7306 | 262.6764 | 8.96% |
| LSTM | 6.2424 | 51.5547 | 3.89% |

The result presented from Table I show the performance of three different models on forecasting tasks which is evaluated using MAE, MSE, and MAPE . The proposed Method which is the hybrid model outperformed the other model.

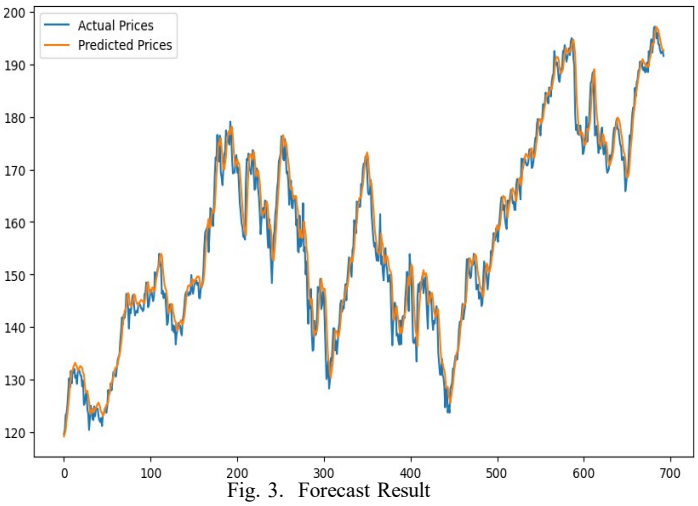

Fig. 3. Forecast Result

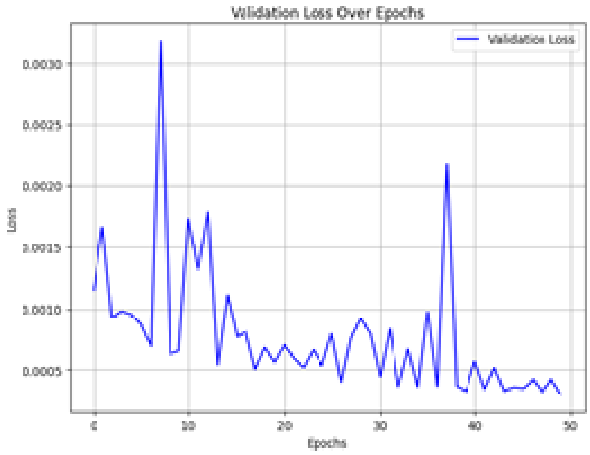

Fig. 4. Validation Lost Graph

The graphic from Fig. 3 shows the results of forecasting Apple stock prices over a given time frame. The blue line represents the actual input of the stock prices, while the orange line represents the predicted stock prices. The graphic from Fig. 4 shows the validation loss of the model over 50 training epochs, which measures how well the performance of the model on an unseen data. Initially, the validation loss decrease demonstrating that the model is learning and improving. The strong overlap between the blue and orange lines in Fig. 3 highlights the model's effectiveness in capturing the key patterns and trends in stock price data. The small difference between the actual and predicted values reflect a high degree of correlation, demonstrating the accuracy and reliability of the forecasting model.

The result of this study shows that the proposed model significantly outperforms the Mini Transformer and LSTM models across all evaluation metrics. With a MAE of 2.4847, MSE of 10.4762, and MAPE of 1.61%, the proposed model demonstrates superior predictive accuracy compared to the Mini Transformer and LSTM. However, whule the model performs well in modeling gradual pries movements, it occasionally struggles with sudden fluctuations, indicating some limitations in handling extreme market events.

The findings of this study have significant implications for both academic research and practical use. The enhanced performance of the hybrid model reinforces the increasing evidence that merging self-attention mechanisms with recurrent architectures can improve forecasting abilities in complex, noisy financial settings. This indicates that investment firms, traders, and financial analysts might utilize such hybrid models to enhance forecasting precision and develop more effective risk management strategies. However, the model's struggle to predict sudden price fluctuations highlights the necessity for future improvements, such as incorporating external signals (like news sentiment and macroeconomic indicators) or creating methods to better manage structural changes. Tackling these issues could enhance the model's resilience and adaptability, making it more suitable for real-world financial decision-making in volatile conditions.

## V. Conclusion

In this study, the researcher explored three different approaches to predict the Apple Inc.'s stock prices, which using a Mini Transformer, LSTM, and a hybrid model

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combining both architectures. These architecture then is assessed using MAE, MSE, and MAPE metrics, this showed that these models effectively capture the complex patterns in financial time series data. The results of this research highlight the strong predictive performance of the proposed method which is a hybrid model, combining Mini Transformer and LSTM architecture in capturing patterns in sequential data.

Future research could enhance these results by exploring deeper or pre-trained Transformer architectures such as Longformer or TiDE, incorporating external features like news sentiment or macroeconomic indicators, and applying advanced optimization techniques such as learning rate scheduling, attention regularization, or neural architecture search (NAS). Furthermore, explainability methods such as SHAP or attention-based visualizations could be integrated to improve model interpretability for financial decision-making.

## AUTHOR'S CONTRIBUTION

R.H. develop the study and was the primary author. Z.D.S carried out the experimental work and helped write the publication. While A.A.G.S monitor the research and provide continuous guidance throughout the study. All authors have thoroughly reviewed and given their approval for the final version of the manuscript.

## DATA'S AVAILABILITY

The historical stock data from Apple Inc. used in this study is obtained using the yfinance library from Python which is easy to access from Yahoo Finance. The data is available on https://finance.yahoo.com/quote/AAPL/history/.

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